Homework #3

3-1. Consider the circuit of Fig.3.1.

a) Using the simple model with $V_{Don} = 0.7V$, solve for I_D;

b) Find I_D and V_D using the ideal diode equation. Use $I_S = 10^{-14}$ A and T=300 K.

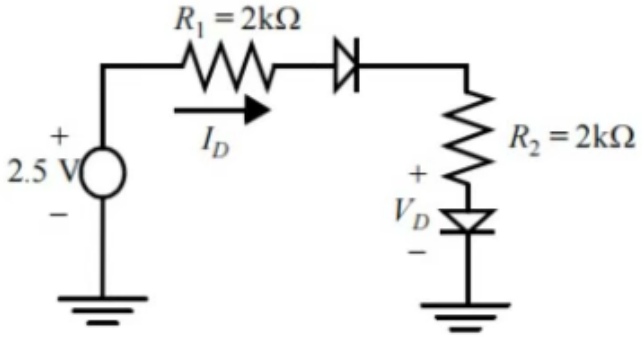
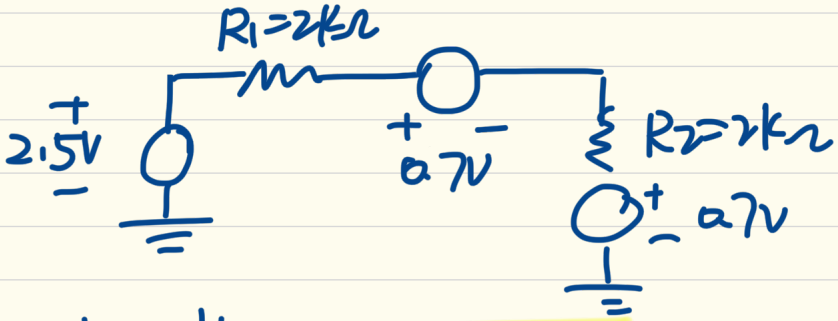


Fig.3.1

解: a)



$$I_D = \frac{V_S - 2V_{D,on}}{R_1+R_2} = 0.275mA$$

$T = 300k$

$I_S = 10^{-14}A$

$$\phi_T = \frac{kT}{q} = 26mV$$

b) $I_D = I_S(e^{\frac{V_D}{\phi_T}} - 1)$

$2V_D + 2I_DR_S = V_S$

$$\Rightarrow 2V_D + 2I_SR_S(e^{\frac{V_D}{\phi_T}} - 1) = V_S$$

$$2V_D + 4\times10^{-11}(e^{\frac{V_D}{26\times10^{-3}}} - 1) = 2.5$$

解得 $V_D = 0.628V$, 故 $I_D = 0.309mA$

3-2. For the circuit in Fig.1.2, V_s=3.3 V. Assume $A_D = 12\ \mu m^2$, $\varphi_0 = 0.65$ V, and m= 0.5, $N_A = 2.5\times10^{16}\ cm^{-3}$, $N_D = 5\times10^{15}\ cm^{-3}$ and $\varepsilon_{si} = 11.7\varepsilon_0$, $\varepsilon_0 = 8.85\times10^{-12} F/m$.

a) Is the diode forward- or reverse-biased?

b) Find I_D and V_D;

c) Find the depletion region width, W_j, of the diode;

d) Use the parallel-plate model to find the junction capacitance, C_j;

e) Set V_s = 1.5 V. Again using the parallel-plate model, explain qualitatively why C_j increases.

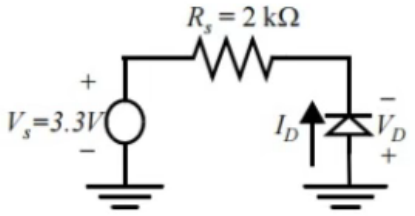


Fig.3.2

解. a) reverse-biased

$\frac{N_A+N_D}{N_AN_D} = 2.4\times10^{-22}\ m^3$

b) $I_D = 0A$, $V_D = -3.3V$

c) $W_j = \sqrt{\left(\frac{2\varepsilon_i}{q}\frac{N_A+N_D}{N_AN_D}\right)(\phi_0 - V_D)} = 1.108\times10^{-6}\ m$

d) $C_{j0} = A_D\sqrt{\left(\frac{\varepsilon_{si}q}{2}\frac{N_AN_D}{N_A+N_D}\right)\phi_0^{-1}} = 2.765fF$

$C_j = \frac{C_{j0}}{(1-V_D/\phi_0)^m} = 1.122fF$

e) $V_D = -V_s$.

$V_s\downarrow, V_D\uparrow, C_j = \frac{C_{j0}}{(1-\frac{V_D}{\phi_0})^m}\uparrow$

电压减小，耗尽层宽度W_j减小，电容C_j增大

3-3. Fig.3.3 shows NMOS and PMOS devices with drains, source, and gate ports annotated. Determine the operation region (saturation, linear, or cutoff) and the drain current I_D for each of the biasing configurations given in table. Assume the model parameters from Table.3.1, V_{BS}=0 and W/L = 1, L=1μm fill the table

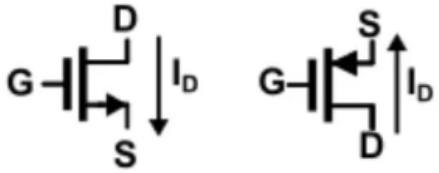


Fig.3.3

	V_{GS}(V)	V_{DS}(V)	Operation region	I_D(μA)
NMOS	2.5	2.5	Saturation	392.04
	3.3	2.2	Triode	726
	0.6	0.1	Cut-off	0
PMOS	-0.5	-1.25	Cut-off	0
	-2.5	-1.8	Triode	176.58
	-2.5	-0.7	Saturation	101.5

NMOS: ① $V_{GS}=2.5V$, $V_{DS}=2.5V$,

$K'=\frac{1}{2}\mu C_{ox}=110\mu A/V^2$

$V_{DS}>V_{GS}-V_{TH}$, 工作在饱和区

$\lambda=0.04V^{-1}$

$I_D=\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2(1+\lambda V_{DS})=392.04\mu A$

② $V_{GS}=3.3V$, $V_{DS}=2.2V$

$V_{DS}<V_{GS}-V_{TH}$, 工作在三极管区

$I_D=\mu_n C_{ox}\frac{W}{L}[(V_{GS}-V_{TH})V_{DS}-\frac{1}{2}V_{DS}^2]=726\mu A$

③ $V_{GS}=0.6V$.

$V_{GS}<V_{TH}$, 工作在夹断区, $I_D\approx 0$

PMOS: ① $V_{GS}=-0.5V$

$K'=\frac{1}{2}\mu C_{ox}=50\mu A/V^2$

$\lambda=0.05V^{-1}$

$|V_{GS}|<|V_{TH}|$，工作在截止区，$I_D\approx 0$

② $V_{GS}=-2.5V$，$V_{DS}=-1.8V$

$|V_{DS}|=|V_{GS}|-|V_{TH}|$，工作在饱和区

$I_D=\frac{1}{2}\mu_p C_{ox}\frac{W}{L}(|V_{GS}|-|V_{TH}|)^2(1+\lambda|V_{DS}|)=176.58\mu A$

③ $V_{GS}=-2.5V$，$V_{DS}=-0.7V$

$|V_{DS}|<|V_{GS}|-|V_{TH}|$，工作在三极管区

$I_D=\mu_p C_{ox}\frac{W}{L}[(|V_{GS}|-|V_{TH}|)|V_{DS}|-\frac{1}{2}V_{DS}^2]=101.5\mu A$

3-4. An NMOS device is plugged into the test configuration shown below in Fig .3.4 The input V_{in} is 2V. The current source draws a constant current of 50 μA. R is a variable resistor between 10kΩ and 30 kΩ. Transistor M1 has following transistor parameters: k'= 110μA/V^2, V_T = 0.7V, and V_{DSAT} = 0.6V, and has a W/L = 2.5μ/0.25. For simplicity, the body effect and channel length modulation can be neglected, i.e λ=0, γ=0.

a) When R =10kΩ find the operation region, V_D and V_S.

b) For the case of R = 10kΩ, would V_S increase or decrease if $\lambda \neq 0$. Explain qualitatively.

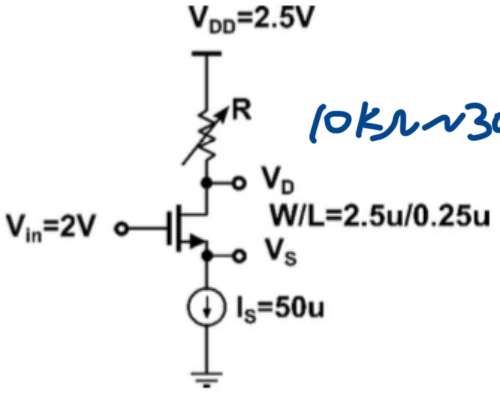


Fig.3.4

K'=110μA/V²
VT=0.7V
VDSAT=0.6V
10kΩ~30kΩ

解. a) R=10kΩ时, $V_D = V_{DD} - I_S R = 2V$

$V_D > V_{DSAT}$, Saturation Region

$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2 = 50\mu A$

$\Rightarrow V_{GS} = 0.913V \Rightarrow V_S = 1.087V$

b) $I_D' = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2(1+\lambda V_{DS})$

故 V_{GS} ↓, V_S ↑

3-5. Show that two MOS transistors connected in parallel with channel widths of W_1 and W_2 and identical channel lengths of L can be modeled as one equivalent MOS transistor whose width is W_1+W_2 and whose length is L, as shown in Fig.3.5 Assume the transistors are identical except for their channel widths.

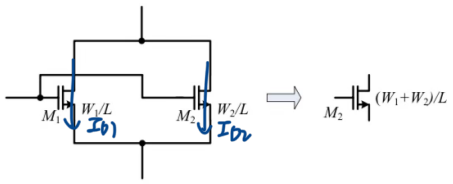


Fig.3.5

解: ① Cutoff, $I_{D1}=I_{D2}=0$

② Triode, $I_{D1}=\frac{1}{2}\mu_n C_{ox}\frac{W_1}{L}[(V_{GS}-V_{TH})V_{DS}-\frac{1}{2}V_{DS}^2]$

$I_{D2}=\frac{1}{2}\mu_n C_{ox}\frac{W_2}{L}[(V_{GS}-V_{TH})V_{DS}-\frac{1}{2}V_{DS}^2]$

$I=I_{D1}+I_{D2}=\frac{1}{2}\mu_n C_{ox}\frac{W_1+W_2}{L}[(V_{GS}-V_{TH})V_{DS}-\frac{1}{2}V_{DS}^2]$

令 $W'=W_1+W_2$

③ Saturation 同②, $W'=W_1+W_2$

3-6. Show that two MOS transistors connected in series with channel lengths of L_1 and L_2 and identical channel widths of W can be modeled as one equivalent MOS transistor whose width is W and whose length is L_1+L_2, as shown in Fig. 3.6. Assume the transistors are identical except for their channel lengths. Ignore the body effect and channel-length modulation.

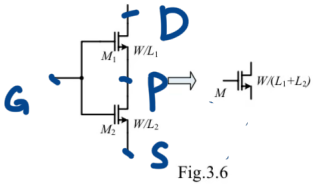


Fig.3.6

解. ① Cutoff, $I_D=0$

② Triode, $I_{D1}=\frac{1}{2}\mu_n C_{ox}\frac{W}{L_1}[(V_{GP}-V_{TH})V_{DP}-\frac{1}{2}V_{DP}^2]$

$I_{D2}=\frac{1}{2}\mu_n C_{ox}\frac{W}{L_2}[(V_{GS}-V_{TH})V_{PS}-\frac{1}{2}V_{PS}^2]$

由 $I_D=I_{D1}=I_{D2}$

$$\begin{cases} L_1 I_D=\frac{1}{2}\mu_n C_{ox}W[(V_{GP}-V_{TH})V_{DP}-\frac{1}{2}V_{DP}^2] \\ L_2 I_D=\frac{1}{2}\mu_n C_{ox}W[(V_{GS}-V_{TH})V_{PS}-\frac{1}{2}V_{PS}^2] \end{cases}$$

$$(L_1+L_2)I_D=\frac{1}{2}\mu_n C_{ox}W\left[(V_{GP}-V_{TH})V_{DP}-\frac{1}{2}V_{DP}^2+(V_{GS}-V_{TH})V_{PS}-\frac{1}{2}V_{PS}^2\right]$$

$$(V_{GP}-V_{TH})V_{DP}-\frac{1}{2}V_{DP}^2+(V_{GS}-V_{TH})V_{PS}-\frac{1}{2}V_{PS}^2$$

$$=-V_{TH}V_{DS}+V_{GP}V_{DP}+V_{GS}V_{PS}-\frac{1}{2}V_{DP}^2-\frac{1}{2}V_{PS}^2$$

$$(V_{GS}-V_{TH})V_{DS}-\frac{1}{2}V_{DS}^2$$

$$=-V_{TH}V_{DS}+(V_{GP}+V_{PS})(V_{DP}+V_{PS})-\frac{1}{2}(V_{DP}+V_{PS})^2$$

$$=-V_{TH}V_{DS}+V_{GP}V_{DP}+V_{GS}V_{PS}+V_{DP}V_{PS}-\frac{1}{2}V_{DP}^2-\frac{1}{2}V_{PS}^2-V_{DP}\cdot V_{PS}$$

$$=-V_{TH}V_{DS}+V_{GP}V_{DP}+V_{GS}V_{PS}-\frac{1}{2}V_{DP}^2-\frac{1}{2}V_{PS}^2$$

故 $I_D=\frac{1}{2}\mu_n C_{ox}\frac{W}{L_1+L_2}\left[(V_{GS}-V_{TH})V_{DS}-\frac{1}{2}V_{DS}^2\right]$

$L'=L_1+L_2$

③ 同②，　　$L'=L_1+L_2$